

Jack Birkin, Ph.D.

AI Engineer

Email: jbirkin10@gmail.com | Phone: (979)-326-5760

LinkedIn: <https://www.linkedin.com/in/jack-birkin/> | Website: jbirkin.github.io

Professional Experience

AI Engineer

Famêlia | Remote | Feb 2026 – Present

- Architected core LLM components within a LangGraph-based agentic pipeline (FastAPI on Azure, Azure AI Foundry) generating personalized recipe substitutions for families with conflicting dietary restrictions.
- Cut end-to-end latency 75% ($\sim 50s \rightarrow \sim 12s$) through parallelization, request batching and caching guided by node-level profiling, and routed structured extraction to GPT-4o while sending user-facing prose to Claude Sonnet — Claude-quality output at $\sim 63\%$ lower latency than an all-Claude baseline.
- Built a retrieval-augmented generation system end-to-end over a 3,500-page clinical nutrition reference: table-aware PDF ingestion and outline-based chunking into 1,844 chunks, Azure OpenAI embeddings, and pgvector/HNSW search in Supabase, serving a dietitian chatbot as grounded retrieval rather than prompt-stuffing.
- Introduced production engineering practice as the sole AI engineer: pytest gates on every deploy workflow (342 hermetic tests), end-to-end LangSmith tracing, and an offline evaluation harness replaying captured production cases to diff graded scores across commits.
- Implemented governance guardrails on LLM output, including allergen/restriction compliance scoring, automated validation against medical conditions, and human-in-the-loop dietitian approval before patient delivery.

Postdoctoral Research Associate

Texas A&M University, College Station, TX (Remote) | Nov 2022 – Feb 2026

- Applied MCMC-based Bayesian modeling to noisy observational data to extract robust source parameters, supporting proposals totaling 800+ observation hours and contributing to 25 peer-reviewed publications (900+ citations).
- Constructed a Python pipeline aggregating low-SNR data across many sources, improving weak-feature detection 4x via four transformation methods, with bootstrap resampling and BIC model selection to guard against overfitting.
- Developed [cubespecfit](#), an open-source Python package for joint modeling of 3-D spatial-spectral data, and a Dash/Python [web application](#) on AWS for collaborative team-wide data review.

Graduate Researcher

Durham University, UK | Oct 2018 – Aug 2022

- Led cleaning and analysis of a 2 TB dataset, applying PCA-based background subtraction and stacking to recover faint signals.
- Analyzed billion-element simulations (~ 5 TB) with inverse modeling to recover signals from initial conditions, cross-validated against paired simulation datasets.

Skills

Software: Python, SQL, PostgreSQL, LangGraph, Git, CI/CD, FastAPI, TypeScript, AWS, Azure, Linux, pandas, Dash

AI / LLMs: Generative AI, prompt engineering, RAG, multi-model pipeline design, LLM orchestration, Azure AI Foundry, knowledge graphs, agentic AI, Claude Code

Data Science / Machine Learning: Regression, spectral data modeling, predictive modeling, data transformation, feature engineering, anomaly detection, model selection (BIC/AIC), uncertainty quantification, bootstrap resampling, MCMC

Leadership: Problem solving, project design and leadership, mentoring, presenting and communicating technical concepts

Education

Ph.D., Physics | Durham University, UK | Oct 2018 – Aug 2022

MPhys, Physics & Astronomy (First Class Honours) | Durham University, UK | Oct 2014 – Jun 2018

Certifications

AWS Certified Solutions Architect - Associate | Amazon Web Services

AWS Certified Cloud Practitioner | Amazon Web Services

Machine Learning Specialization | Stanford University & DeepLearning.AI